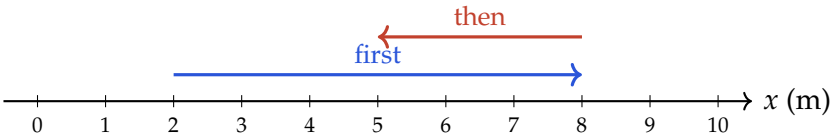
AP PHYSICS 1

Unit 1 • Lesson 1.1 • Guided notes

Topic 1.1 • Scalars and Vectors in One Dimension

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: tell scalars from vectors, and use signs to give the direction of a vector in one dimension.

Scalar or vector?	A _____ has a magnitude only (a number with a unit). A _____ has a magnitude **and** a _____. **Scalars**: _____, _____, time, mass **Vectors**: _____, _____, acceleration, force
Direction in one dimension	On a line, we choose a _____ direction (for example: right, east, or up). A vector pointing that way has a _____ sign; the opposite way, a _____ sign. The sign of a vector gives its _____, not its size: $-8\,\text{m/s}$ is _____ than $+3\,\text{m/s}$.
Example: distance vs. displacement	 A cart moves from $x = 2\,\text{m}$ to $x = 8\,\text{m}$, then back to $x = 5\,\text{m}$. Distance = _____ Displacement $\Delta x = x_f - x_i =$ _____ Distance is the total _____ (never negative). Displacement is the _____: it only depends on the start and the end.
Adding vectors in one dimension	Add the signed values. A walker goes $+5\,\text{m}$ then $-7\,\text{m}$: total displacement = _____, distance = _____. Two forces on a box: $+30\,\text{N}$ and $-12\,\text{N}$. Net force = _____ (to the right).
Check	A runner does one full lap of a 400 m track and stops at the start line. Distance = _____ Displacement = _____
Summary	